

# Data-Driven Clustering of MLB Swing Mechanics Using High-Frequency Motion Tracking Data

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**Abstract**—The characterization of baseball batters have long used clustering techniques on biomechanical swing metrics at discrete moments, but such efforts are sparse and have yet to leverage complete swing trajectory data that could explain more granular differences between batters’ swings, better enhancing player development strategies. We propose a pipeline that uses the following techniques to quantify differences between batter profiles. To model continuous data, Functional Data Analysis (FDA) is used to extract swing information and represent it in a lower dimension. Uniform Manifold Approximation and Projection (UMAP) is then used to uncover nonlinear structure on swing embeddings generated from Functional Principal Component Analysis (FPCA). The combination of Gaussian mixture models and hierarchical clustering assigns batters to four clusters with associated measures of uncertainty. The profiling of clusters rely on ranked feature importance generated from random forests. In this study, we find that four distinct batter profiles emerge, with the primary differences between them being related to timing, lower-body behavior, rotational patterns, and actions at the point of contact.

## I. INTRODUCTION

Hitting a baseball is one of the most mechanically complex actions in sports. A successful swing depends not only on strength and bat speed but also on precise timing, coordination, and efficient sequencing of body segments. With the rise of high-resolution motion capture systems like Hawk-Eye, teams now have access to thousands of joint trajectories per swing, offering a window into the biomechanics of each hitter’s motion.

Until now, much of the research in baseball biomechanics has focused on identifying and correlating specific swing metrics—such as hip-shoulder separation, bat angle, and center-of-mass positioning—with performance outcomes like exit velocity, launch angle, and swing-and-miss rate [4], [7]. These studies typically rely on engineered features at discrete moments in the swing (e.g., at contact or heel strike), often assuming predefined groupings of players. While such work has helped identify important markers of performance—such as lead leg force generation, bat path steepness, and angular velocities of the torso and arms [5]—it falls short of capturing the continuous and coordinated nature of full-body motion.

However, despite the abundance of detailed joint tracking data, few studies have explored how hitters might be grouped based on their entire 3D movement trajectories. Existing work on swing clustering has generally used summary metrics like bat speed or swing time to categorize swing styles [1]. Some

studies applied k-means clustering to these features and identified meaningful swing types such as “fast and long” versus “slow and short” [1]. Yet these efforts do not leverage the full richness of the time-series biomechanical data. Functional Data Analysis (FDA) offers a more principled framework by modeling joint trajectories as smooth curves and decomposing those curves using Functional Principal Component Analysis (FPCA), a technique that has been widely used in gait, running, and jumping biomechanics [3], [6]. Surprisingly, FPCA has seen little application in baseball swing analysis, despite its potential to capture entire motion patterns and discover interpretable modes of variation between players.

Therefore, this study aims to identify data-driven mechanical archetypes among MLB hitters by modeling their swings as continuous functions and clustering them based on full-body 3D biomechanics. Specifically, we ask: Can we discover distinct swing types directly from motion data, and how do those swing types relate to meaningful outcomes like exit velocity, bat speed, and contact quality?

We hypothesize that variation in upper-body joint angles—particularly those involving the wrists, elbows, and shoulders—will serve as primary differentiators of swing mechanics. While lower-body movements tend to be more uniform across players, upper-body motion often reflects individualized techniques that can influence swing effectiveness. Furthermore, we focus on biomechanically meaningful positional and kinematic features that correspond to common coaching concepts, such as hand path adjustments, torso tilt, stride length, and weight transfer. Rather than relying solely on raw spatial coordinates, our analysis emphasizes movement patterns that are directly tied to how hitters load, sequence, and deliver the bat through the swing.

The objective of this paper is to use functional representations of swing data to uncover interpretable, biomechanically meaningful clusters of hitters and evaluate how those clusters relate to performance. Our broader goal is to help teams move beyond snapshot metrics and toward a holistic, movement-based understanding of what makes a swing effective.

## II. METHODS

This study leverages a high-resolution, full-body motion capture dataset collected at 300 frames per second from Major League stadiums during live games. For each swing, the system tracks over 70 biomechanical variables, including the

3D positions of key joints (e.g., wrists, elbows, shoulders, hips, knees, and ankles) as well as joint angles representing rotation, flexion, and separation across the upper and lower body. The data include both raw spatial coordinates and derived angular measurements that capture body orientation and sequencing throughout the swing. Only right-handed batters facing fastballs located near the center of the strike zone were considered in this analysis to minimize confounding variation due to pitch type or handedness. In total, the dataset comprises 100 right-handed professional hitters, each with 10 recorded swings.

In addition to raw kinematic data, we also use a companion dataset that summarizes swing characteristics and outcomes for each trial. These summary features include bat speed, attack angles, contact location, and time-stamped biomechanical events (e.g., peak gather, heel strike, and bat-ball contact). Together, the two datasets offer a complete temporal and mechanical view of each swing—from preparatory movements to post-contact metrics.

To analyze this data, we developed a multi-stage pipeline consisting of: (1) temporal alignment of swing phases; (2) joint selection using dimensionality reduction; (3) functional data representation of joint trajectories using B-splines; (4) dimensionality reduction via FPCA; (5) nonlinear embedding using UMAP; (6) clustering with Gaussian Mixture Models (GMM); (7) creating robust batter-level clusters with consensus clustering; and (8) qualitatively profiling clusters using informative features from the summary dataset. To ensure stability in cluster assignments, we employed consensus clustering across 250 randomized UMAP+GMM iterations. This pipeline allowed us to identify consistent biomechanical patterns among hitters and group them into distinct swing archetypes based on full-body motion.

#### A. Swing Phase Isolation and Temporal Normalization

We focused on the pre-contact swing phase, defined as the interval between the peak gather time (when the hips begin to move forward) and the estimated moment of bat-ball contact. This window represents the most biomechanically explosive segment of the swing, where kinematic differences are most likely to influence outcomes.

Due to natural variations in swing duration, we rescaled the time within this window to the unit interval  $[0, 1]$  for each swing. This normalized time, denoted as  $t$ , allowed us to compare equivalent motion phases across swings:

$$t = \frac{\text{Frame} - \text{Frame}_{\text{Gather}}}{\text{Frame}_{\text{Contact}} - \text{Frame}_{\text{Gather}}}, \quad t \in [0, 1] \quad (1)$$

If any key timestamp was missing (e.g., the peak gather time), it was imputed using the player-specific average across their other recorded swings to avoid data loss while preserving swing structure.

#### B. Joint Selection and Dimensionality Reduction

To reduce noise and redundancy in the full set of tracked markers, we identified a subset of joint variables that capture the most informative movement patterns across the body. Rather than using all three Cartesian coordinates (X, Y, Z) per joint, we applied principal component analysis (PCA) to each joint's 3D trajectory individually. This allowed us to extract the single axis of maximal variation—the direction along which the joint moves most consistently during the swing.

Specifically, for each tracked joint, we performed PCA on its time series of X, Y, and Z coordinates, retaining only the first principal component (PC1). This dimension represents the dominant movement direction of the joint and reduces the raw 3D trajectory to a one-dimensional curve without losing key biomechanical structure. To account for inter-individual differences in stature, joint Y-coordinates (i.e., vertical position) were normalized by player height prior to PCA. The result is a parsimonious yet interpretable representation of motion, with each joint trajectory reduced to a single functional variable aligned with its primary axis of movement.

This preprocessing step yielded a concise set of joint curves for downstream functional analysis. These selected motion paths were then smoothed using B-splines and analyzed using multivariate FPCA.

#### C. Functional Representation of Joint Trajectories

To enable principled comparisons of biomechanical patterns across hitters, we modeled joint motion using Functional Data Analysis (FDA). Specifically, the time-varying trajectory of each selected joint was expressed as a smooth curve over normalized swing time. Each trajectory  $f_i(t)$  was approximated using a linear combination of cubic B-spline basis functions:

$$f_i(t) = \sum_{k=1}^K c_k B_k(t) \quad (2)$$

where  $B_k(t)$  is the  $k$ -th basis function and  $c_k$  is its corresponding coefficient. We used  $K = 12$  basis functions, which balances flexibility with smoothness and prevents overfitting to noisy frame-level data. This transformation yielded a compact set of coefficients per joint per swing, encoding the full shape and timing of movement prior to bat-ball contact.

Figure 1 illustrates the smoothing process for a representative joint trajectory. The raw motion trace, sampled at 300 Hz, exhibits frame-to-frame jaggedness due to measurement granularity and microvariations. The fitted B-spline captures the underlying biomechanical pattern in a continuous, noise-robust form. These functional representations served as the foundation for all downstream analyses including FPCA and clustering.

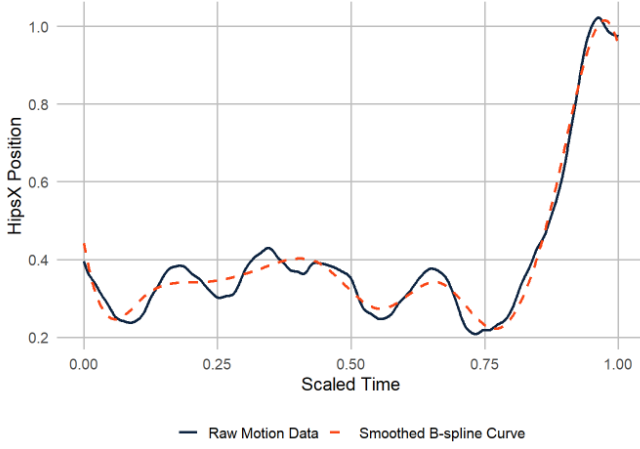


Fig. 1: Example of B-spline smoothing applied to a swing trajectory. The raw position data (solid line) is smoothed using a cubic B-spline (dashed line), yielding a continuous curve over scaled time. This enables functional representation of movement prior to bat-ball contact.

#### D. Multivariate Functional Principal Component Analysis

To uncover dominant movement patterns across swings, we performed FPCA on the set of smoothed functional curves representing each joint's motion. This approach captures variation in full trajectory shapes across joints and swings, preserving temporal dependencies. For a joint curve  $f_i(t)$  from swing  $i$ , FPCA seeks eigenfunctions  $\phi_k(t)$  that solve:

$$\max_{\phi_k} \text{Var} \left( \int f_i(t) \phi_k(t) dt \right) \quad (3)$$

The integral computes the inner product between the joint curve and eigenfunction  $\phi_k(t)$ , yielding a score that represents how much the swing expresses the  $k$ -th mode of variation. Each swing is thus represented by a vector of functional principal component scores that summarize the shape and coordination of joint trajectories.

#### E. Nonlinear Embedding via UMAP

Although FPCA reveals interpretable linear variation, we observed limited separation between swings in the resulting score space. To capture nonlinear structure, we embedded swings using Uniform Manifold Approximation and Projection (UMAP). UMAP constructs a weighted graph in the FPCA space, where edge weights  $p_{ij}$  quantify local similarity using an exponential kernel:

$$p_{ij} = \exp \left( -\frac{\|z_i - z_j\|^2}{\sigma_i} \right) \quad (4)$$

Here,  $z_i$  and  $z_j$  are the FPCA score vectors for swings  $i$  and  $j$ , summarizing their full-body movement trajectories, and  $\sigma_i$  is a local distance scaling parameter that adapts to the density around swing  $i$ . UMAP then optimizes a low-dimensional embedding by minimizing the cross-entropy between the high- and low-dimensional graphs. In the resulting 2D or 3D embedding, nearby points represent swings with similar biomechanical patterns.

#### F. Clustering with Gaussian Mixture Models

To identify distinct swing archetypes, we clustered swings in the UMAP-embedded space using GMM. Let  $x_i \in \mathbb{R}^3$  denote the UMAP embedding for swing  $i$  in a 3-dimensional embedding. GMM assumes that these embedded vectors are drawn from a mixture of  $K$  multivariate Gaussian distributions:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k) \quad (5)$$

where  $\pi_k$  is the prior weight,  $\mu_k$  is the mean, and  $\Sigma_k$  is the covariance matrix of the  $k$ -th cluster. Parameters were estimated via Expectation-Maximization (EM), and the optimal number of clusters were selected by minimizing the Bayesian Information Criterion (BIC), but these resulted in an unreasonably high number of clusters which was much too granular for any type of generalization, leading us to make the decision to choose the number of clusters with a visual analysis. The probabilistic nature of GMM allowed us to assess confidence in swing-to-cluster assignments.

We then evaluated within-player consistency by quantifying how many of a batter's ten swings fell into the same cluster. This swing-level consistency score offered insight into each player's mechanical repeatability.

#### G. Batter-Level Clustering and Consensus Assignment

To create a single biomechanical profile per batter, we averaged each batter's FPCA scores across their ten swings, producing a 42-dimensional vector (3 components per 14 joints). We chose to average scores in the FPCA space rather than average the original smoothed joint curves and re-run FPCA because this approach retains information about within-batter variability. Averaging in the original curve space would artificially smooth out meaningful differences between swings, whereas averaging FPCA scores provides a compact summary of how each batter tends to vary along key biomechanical modes. We then repeated UMAP and GMM clustering on these batter-level summaries. However, due to UMAP's stochasticity, different runs could yield different cluster structures.

To address this problem, we implemented consensus clustering. This involved running UMAP and GMM 250 times with varying random seeds. For each run, we recorded the cluster assignment of each batter and computed a co-occurrence matrix, where each entry counts how often a pair of batters appeared in the same cluster. To align cluster labels (such that, for example, a batter's assignment to Cluster 1 was a practical assignment to Cluster 1 across all iterations, as what "Cluster 1" may be could change every iteration), This matrix was converted to a dissimilarity matrix and subjected to hierarchical clustering to derive final robust clusters. Cluster robustness was quantified as the average co-clustering frequency among members. This approach allowed us to balance the flexibility of nonlinear embedding with the reproducibility necessary for actionable biomechanical classification.

### H. Cluster Profiling

To interpret and validate the batter-level clusters obtained from consensus clustering, we conduct a post hoc analysis using the summary biomechanical dataset. To further ensure higher confidence in the profiles described, we derived a representative set of batters for each cluster. An incentive behind creating such representative sets are to compensate for the fact that we are using hard class assignments instead of leveraging the cluster assignment probability distribution of each batter (which would provide more nuance). A batter qualifies as a part of the representative set if it meets a consistency threshold ( $n\%$ ), meaning it's assigned to its respective cluster in over  $n\%$  of the 250 iterations of UMAP+GMM. This would indicate a batter with stable cluster membership.

Each batter's final cluster assignment serves as the target variable in a random forest model trained on this summary data. This approach offers two key advantages. First, it enables cluster characterization using features that were not involved in the clustering pipeline, thereby reducing circularity and potential bias. Second, the resulting model provides feature importance scores, which highlight the most distinguishing biomechanical attributes associated with each cluster. These insights support a more interpretable understanding of the mechanical differences between hitter archetypes.

For each cluster, the features with the highest importance scores were noted. To assess how individual features differentiate the clusters, we generated parallel coordinate plots for each feature that appeared in the top 5 highest-scoring features for at least one cluster across 20 iterations of random forest modeling. These plots display the median observed value of each feature, separated by cluster. Due to the stochastic nature of random forests, feature importance scores may vary, but the set of features remains at a size of 19 to 23 consistent features for different runs of the random forest. Median is used as the measure for central tendency here for two reasons. First, the representative sets for each cluster are small and thus prone to outliers. Second, a visual analysis of the distributions for these measures show skewness and thus a lack of suitability to use the mean. Uncertainty is quantified with the interquartile range.

Holistically synthesizing these plots for each feature allows us to use domain knowledge to qualitatively describe batter profiles.

## III. RESULTS

### A. Functional Principal Components Capture Key Modes of Variation

After smoothing each swing's joint trajectories into functional curves, we applied FPCA to uncover the dominant patterns of biomechanical variation across swings. FPCA treats entire movement trajectories as structured inputs, allowing us to extract interpretable directions of variation in curve space rather than analyzing static or frame-based features.

The first three functional principal components (FPCs) together explained approximately 82% of the total variance

in joint trajectories. Beyond the third component, additional FPCs contributed marginal explanatory power, indicating that most variation in swing mechanics can be effectively summarized in a low-dimensional subspace.

To further examine whether swing variation could be characterized using linear components alone, we plotted each swing in the space defined by its first three functional principal component scores (Figure 2). The resulting distribution showed no discernible structure or separation between swings, suggesting that variation across joint trajectories may lie along more complex, nonlinear manifolds. This observation motivated our use of nonlinear dimensionality reduction methods in the following analysis.

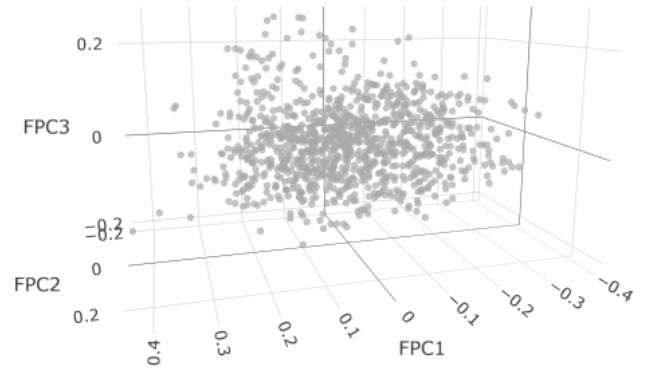


Fig. 2: Scatterplot of swings in FPCA space using the first three functional principal components. The lack of clustering or clear structure suggests that linear components alone may not fully capture the complexity of swing mechanics.

### B. Nonlinear Embedding and Swing-Level Clustering

While FPCA provided a compact linear summary of swing mechanics, the resulting embedding showed no clear structure, suggesting that important biomechanical differences may lie along nonlinear manifolds. To better capture these relationships, we applied UMAP to the swing-level FPCA scores. UMAP constructs a local similarity graph in the high-dimensional space and optimizes a lower-dimensional embedding that preserves neighborhood relationships. The result is a three-dimensional map of 1,000 swings, where spatial proximity reflects biomechanical similarity.

We then applied GMM to identify structure in the UMAP-embedded space. GMM assumes the data are generated from a mixture of multivariate Gaussians and uses expectation-maximization to assign probabilistic cluster labels. Though a six- or seven-cluster solution is optimal according to BIC, a four-cluster solution was selected for interpretability (Figure 3). Figure 5 further corroborates that a four-cluster solution is reasonable, forgoing only a little more dissimilarity than a six-cluster solution would result in.

In Figure 3, we observe that Clusters 1, 2, and 3 form three distinct groups in the UMAP embedding. These clusters separate clearly in the UMAP1 versus UMAP2 space but share relatively constant values along UMAP3, making them

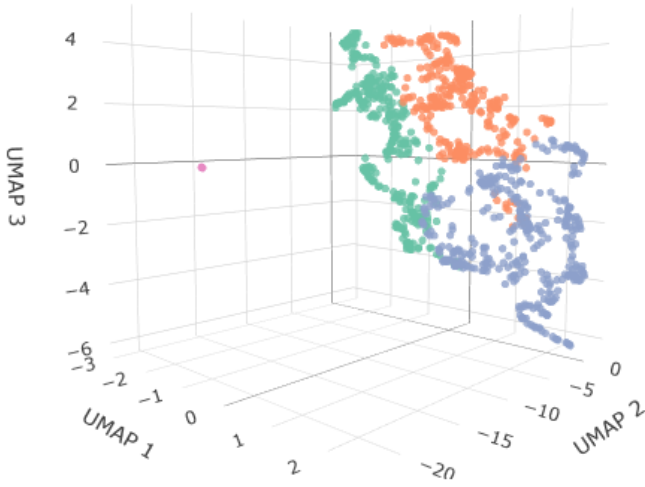


Fig. 3: GMM clustering of swings in UMAP space. Colors represent cluster assignments: Cluster 1 (green), Cluster 2 (orange), Cluster 3 (blue), and Cluster 4 (pink).

appear as planar groupings. Despite this planar appearance, the clusters remain meaningfully distinct based on their local biomechanical features.

Cluster 4, however, is markedly different. It consists solely of Batter 6, whose 10 swings are tightly grouped far from the main plane along the UMAP3 axis, appearing as a distant single point in the embedding. Batter 6 was the only hitter to have *any* of their swings assigned to the swing-level Cluster 4, highlighting their unique swing mechanics relative to the rest of the cohort.

It is important to note that the swing-level clusters and batter-level clusters are not directly equivalent: being assigned to swing-level Cluster 1 does not necessarily imply a batter belongs to batter-level Cluster 1. The swing-level clusters characterize individual swing events, while the batter-level clusters summarize an athlete’s aggregate biomechanical profile across all swings.

To evaluate mechanical consistency, we examined how many swings each batter contributed to each cluster. Batters whose 10 swings all landed in a single cluster can be considered highly repeatable. Table I lists representative batters from each swing cluster—defined as those whose 10 swings were entirely assigned to the same cluster.

| Cluster   | Representative Batters                                                                                             | Count |
|-----------|--------------------------------------------------------------------------------------------------------------------|-------|
| Cluster 1 | 1, 2, 3, 9, 12, 16, 18, 28, 29, 43, 44, 46, 47, 49, 50, 58, 59, 65, 67, 70, 75, 78, 80, 84, 89, 92, 93, 94, 95, 97 | 30    |
| Cluster 2 | 5, 11, 17, 23, 25, 31, 85, 96, 98                                                                                  | 9     |
| Cluster 3 | 8, 19, 32, 39, 63, 72                                                                                              | 6     |
| Cluster 4 | 6                                                                                                                  | 1     |

**TABLE I:** Representative batters for each swing cluster. These players had all 10 swings assigned to the same GMM cluster.

### C. Batter-Level Embedding and Initial Clustering

To identify biomechanical archetypes at the hitter level, we began by aggregating swing-level features into a single representative profile per batter. Specifically, we averaged each player’s functional principal component (FPC) scores across their ten swings, yielding a 42-dimensional vector summarizing their joint motion patterns throughout the swing phase.

These batter-level profiles were then embedded into a three-dimensional space using UMAP, applied to the matrix of averaged FPC scores. UMAP was configured with 10 nearest neighbors and a minimum distance of 0.1, balancing local fidelity and global structure. The resulting embedding organizes hitters based on similarities in full-body biomechanics.

To identify data-driven groups of hitters, we applied GMM to the batter-level UMAP embedding. Each hitter was assigned to the cluster with the highest posterior probability, while soft probabilities were retained to assess uncertainty. We selected a four-cluster solution to align with the swing-level clustering structure, facilitating comparison across analytical levels. This approach enables us to explore how consistent swing patterns aggregate into distinct batter-level archetypes.

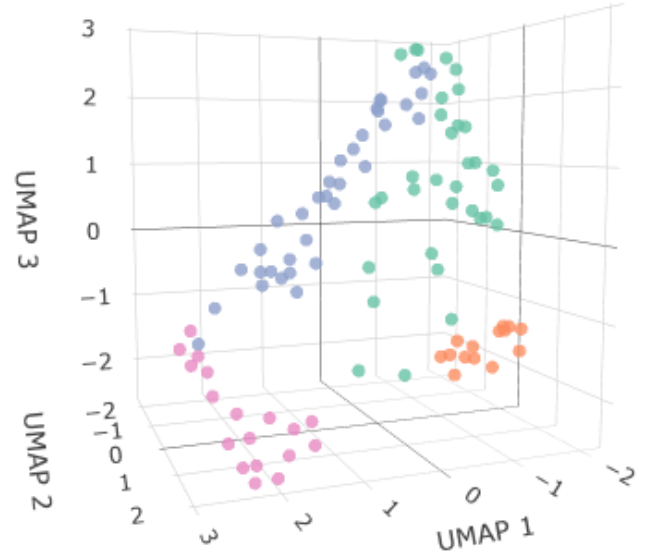


Fig. 4: Initial Gaussian Mixture Model clustering of batters in UMAP space. Each point represents a batter, colored by cluster assignment. Colors: Cluster 1 = green, Cluster 2 = orange, Cluster 3 = blue, Cluster 4 = pink.

### D. Consensus Clustering for Robust Group Assignment

While the initial GMM provided a useful segmentation of hitters, it was sensitive to the inherent randomness in UMAP. Because UMAP is a stochastic algorithm, repeated runs with different random seeds can produce slightly different low-dimensional embeddings, which in turn affect cluster assignments. This poses a challenge when aiming to define stable, reproducible groups of hitters.

Notably, we observed that this variability was more pronounced at the batter level than at the swing level. At the swing level, with 1,000 data points, UMAP consistently converged to similar embeddings across runs. In contrast, the batter-level dataset—comprised of only 100 averaged swing profiles—exhibited higher sensitivity to initialization, leading to less stable cluster boundaries across repeated runs.

To address this, we implemented a consensus clustering approach at the batter level. The procedure aggregates results across multiple iterations of UMAP followed by GMM clustering, thereby reducing dependence on any single run. Specifically, we performed the following steps:

- 1) Repeated UMAP embedding of the 100 batter profiles 250 times, each with a different random seed.
- 2) On each embedding, applied GMM clustering with four components.
- 3) For each pair of batters, counted how often they were assigned to the same cluster across all 250 runs.
- 4) Used the resulting  $100 \times 100$  co-occurrence matrix to define a similarity score between every pair of batters.
- 5) Transformed the co-occurrence matrix into a distance matrix and applied hierarchical clustering using Ward’s linkage to generate the final cluster labels.

This approach allowed us to move from a single stochastic clustering to a robust grouping supported by repeated evidence. Additionally, for each cluster, we calculated a robustness score—defined as the average co-occurrence rate among all batter pairs within that cluster—quantifying how consistently its members were grouped together across runs.

Figure 5 shows the resulting dendrogram colored by consensus cluster. These final groups form the foundation for downstream interpretation and profiling.

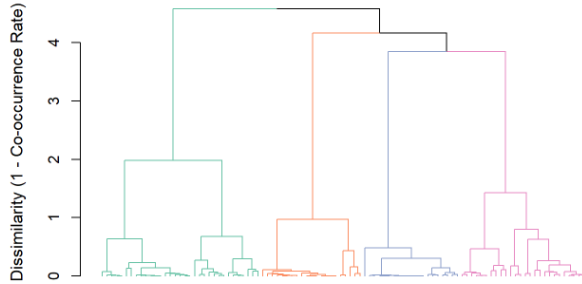


Fig. 5: Hierarchical clustering of batters based on their co-occurrence rates across 250 UMAP+GMM runs. Colored branches indicate final consensus clusters.

To summarize the results of the consensus clustering process, we report the size, membership, and robustness score of each final cluster. Table II lists the four consensus clusters, each containing a distinct set of batters whose swing profiles were consistently grouped together across 250 UMAP+GMM iterations.

Cluster robustness scores are on a scale from 0.00 to 1.00, and the four uncovered clusters had scores ranging from 0.66 to 0.95, indicating strong internal agreement despite the

| Cluster | Batters in Cluster                                                                                                             | Robustness |
|---------|--------------------------------------------------------------------------------------------------------------------------------|------------|
| 1       | 1, 2, 3, 4, 9, 16, 18, 27, 29, 34, 36, 42, 43, 44, 46, 47, 48, 49, 58, 59, 62, 65, 67, 70, 75, 78, 80, 84, 92, 93, 94, 97, 100 | 0.668      |
| 2       | 5, 7, 17, 21, 23, 24, 26, 35, 45, 51, 55, 56, 60, 64, 66, 68, 73, 82, 83, 86, 90                                               | 0.829      |
| 3       | 8, 10, 11, 25, 30, 31, 38, 39, 40, 41, 52, 53, 57, 69, 76, 79, 81, 85, 96, 98                                                  | 0.911      |
| 4       | 6, 12, 13, 14, 15, 19, 20, 22, 28, 32, 33, 37, 50, 54, 61, 63, 71, 72, 74, 77, 87, 88, 89, 91, 95, 99                          | 0.678      |

TABLE II: Summary of final consensus cluster composition based on 250 UMAP+GMM runs including the number of batters and robustness scores quantifying how consistently batters were grouped together across runs.

stochastic nature of the underlying embedding. The highest score was observed in Cluster 3, suggesting especially coherent swing characteristics among its members. Clusters 1 and 4 were larger but slightly less stable, while Cluster 2 combined high robustness with moderate size. These scores provide confidence that the resulting groups reflect meaningful biomechanical archetypes.

#### E. Identifying Reliable Representative Batters for Profiling

Using the clustering results from UMAP+GMM, we derived the consistency threshold to qualify a batter as a part of the representative set using an elbow plot.

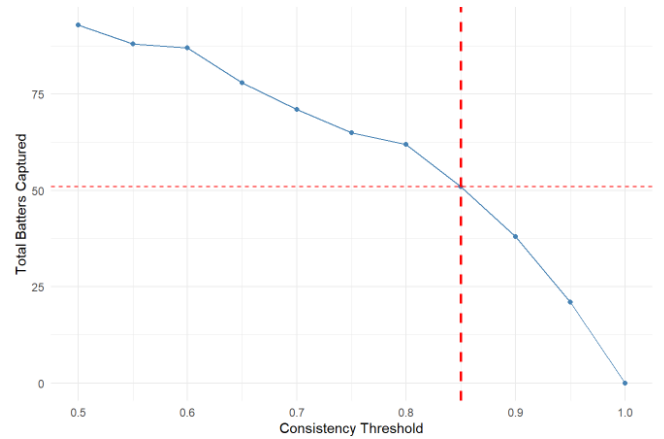


Fig. 6: Elbow plot illustrating the relationship between the consistency threshold and the number of batters captured. The horizontal dashed line represents the inclusion of 55 batters, while the vertical dashed line marks the selected 85% consistency threshold.

We selected an 85% consistency threshold based on where further lowering the threshold produced diminishing returns in the number of representative batters. At this threshold, 55 batters qualify for inclusion, as outlined in Table III. Decreasing the threshold below 85% yielded only marginal gains in the number of included batters while introducing greater uncertainty in cluster assignments. The final representative sets are as follows:



| Cluster | Representative Batters                                                    | Count   |
|---------|---------------------------------------------------------------------------|---------|
| 1       | 1, 3, 9, 16, 43, 46, 49, 59, 67, 75, 80, 84, 93, 94                       | 14 (33) |
| 2       | 5, 7, 17, 21, 23, 24, 26, 35, 45, 56, 60, 64, 66, 68, 83, 90              | 16 (21) |
| 3       | 8, 10, 11, 25, 30, 31, 38, 39, 40, 41, 52, 53, 57, 69, 76, 79, 85, 96, 98 | 19 (19) |
| 4       | 22, 33, 50, 77, 88, 95                                                    | 6 (27)  |

**TABLE III:** Representative batters for each batter cluster. The batters listed were assigned to their respective consensus cluster over 85% of the time in 250 iterations of UMAP and GMM. Numbers in parentheses in the Count column represent the number of batters in the robust cluster denoted in Table III.

#### F. Random Forest Classification Results

To validate the predictive value of biomechanical features, we trained random forest classifiers to predict batter cluster assignments. Models were fit on both the representative subset of batters and the full set of 100 batters, using the summarized biomechanical features as inputs.

1) *Representative Set Performance:* On the representative set, the random forest achieved an out-of-bag (OOB) misclassification rate of 25.5%. The confusion matrix is shown in Table IV.

**TABLE IV:** Random Forest Confusion Matrix on Representative Batters

| True \ Predicted | 1  | 2  | 3  | 4 | Error |
|------------------|----|----|----|---|-------|
| 1                | 13 | 0  | 0  | 0 | 0.00  |
| 2                | 0  | 10 | 5  | 0 | 0.33  |
| 3                | 1  | 4  | 11 | 0 | 0.31  |
| 4                | 3  | 0  | 0  | 4 | 0.43  |

Cluster 1 was classified perfectly, while moderate confusion was observed between Clusters 2 and 3.

2) *Full Dataset Performance:* When training on the full set of 100 batters, the random forest achieved an OOB misclassification rate of 33%. The confusion matrix for the full dataset is shown in Table V.

**TABLE V:** Random Forest Confusion Matrix on All Batters

| True \ Predicted | 1  | 2 | 3  | 4  | Error |
|------------------|----|---|----|----|-------|
| 1                | 27 | 0 | 2  | 4  | 0.18  |
| 2                | 4  | 9 | 5  | 3  | 0.57  |
| 3                | 3  | 3 | 14 | 0  | 0.30  |
| 4                | 8  | 1 | 0  | 17 | 0.35  |

Although the misclassification rate increased slightly when generalizing to the full sample, the performance remained comparable. Cluster 2 consistently showed the highest confusion, which is consistent with its lower robustness observed during consensus clustering.

Overall, random forest classification results suggest that the biomechanical summary features generalize well beyond the representative set. The relatively low error rates support the validity of the clustering structure and indicate that batter-level swing patterns can be reliably distinguished using interpretable features.

#### G. Biomechanical Profiles of Batter Clusters

Following validation of the Random Forest classifier, we leveraged the model's top features to characterize biomechanical differences between clusters. Figure 7 presents parallel coordinate plots that visualize key movement and swing metrics across batter clusters. While not all plotted features were among the highest ranked in feature importance, they were selected for interpretability and biomechanical relevance. The plots encompass timing variables (e.g., time to gather, time to contact), stride parameters (e.g., stride length, stride width), knee flexion angles, hip and torso positioning, hip-shoulder separation, and rotational mechanics (e.g., forward bend, torso rotation). Additionally, we included swing performance indicators (bat speed, attack angles) and player body metrics (height, weight) to contextualize observed movement patterns.

To allow meaningful visual comparisons, all features within each category were normalized to the [0, 1] range. This standardization preserves relative differences between clusters while accommodating variables of differing units and scales. For reference, unscaled median values are reported separately in Table VIII in the Technical Appendix.

These normalized feature patterns revealed distinct mechanical archetypes across the four consensus batter clusters. Despite scaling, clear trends emerged that differentiated hitters in terms of loading mechanics, rotational sequencing, timing strategies, and swing trajectories.

**Cluster 1: Deep Lower-Body Load with Early Bat Entry**  
Batters in this group exhibited strong lower-body engagement, with a median stride length at heel strike of 3.38 ft and deep knee flexion at setup (FrontKneeFlexionSetup = 51.2°, RearKneeFlexionSetup = 61.9°). Hip-shoulder separation was moderate at setup (HSSeparationSetup = -10.0°) and increased positively at contact (HSSeparationContact = +6.0°), indicating efficient rotational sequencing. Forward bend was elevated throughout the swing (ForwardBendSetup = 16.6°, ForwardBendContact = 6.3°), suggesting an athletic posture maintained through impact. Batters entered the hitting zone relatively early (BackZoneDepth = -0.48 ft), with average bat speed (BatSpeedSSMax = 71.0 mph) and strong batted ball metrics (ExitVelocity = 94.1 mph, LaunchAngle = 21°, LaunchDirection = 4.07°). These patterns reflect a swing built around strong preparatory loading, efficient rotation, and early barrel positioning.

**Cluster 2: Upright Setup with Explosive Hand Speed**  
This group demonstrated a more upright stance with reduced knee flexion at setup (FrontKneeFlexionSetup = 34.2°, RearKneeFlexionSetup = 39.3°) and moderate stride length (StrideLengthHeelStrike = 3.30 ft). Hip-shoulder separation at setup was the most pronounced among clusters (HSSeparationSetup = -11.6°), but minimal separation remained at contact (HSSeparationContact = +0.54°), indicating less efficient energy transfer. Forward bend was lower compared to other groups (ForwardBendSetup = 9.3°, ForwardBendContact = 5.7°), supporting the observation of an upright posture. Batters achieved early bat entry (BackZoneDepth = -0.53 ft) and

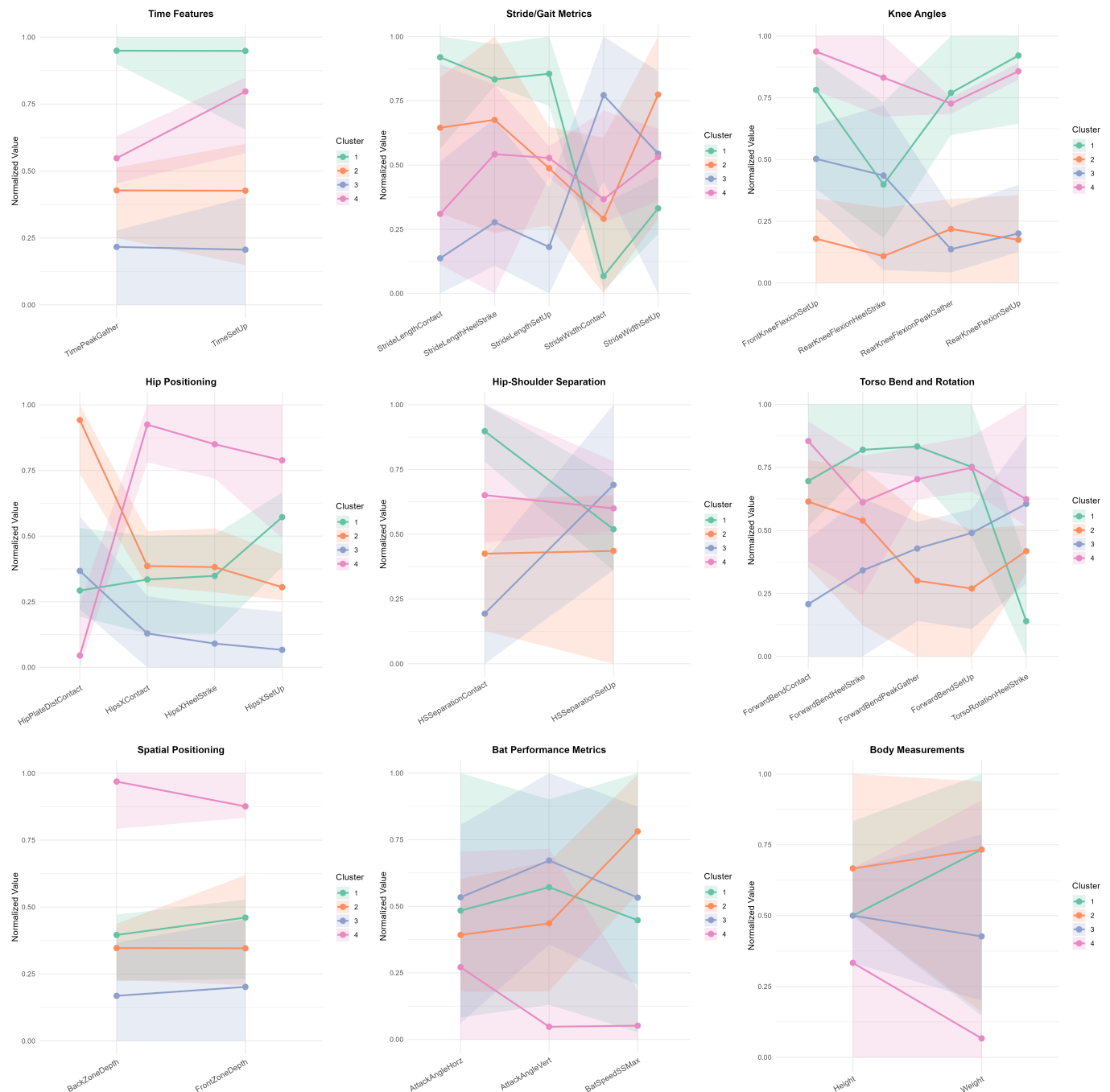


Fig. 7: Standardized median metrics (and IQR bands to measure uncertainty) delineated by feature category and the representative sets of each cluster: Cluster 1 (green), Cluster 2 (orange), Cluster 3 (blue), and Cluster 4 (pink). There are 7 groups of informative features: Time Features, Stride/Gait Metrics, Knee Angles, Hip Positioning, Hip-Shoulder Separation, Torso Bend and Rotation, and Spatial Positioning. Bat Performance Metrics and Body Measurements are included as examples of feature groups that are uninformative of differences between clusters.



the highest median bat speed ( $\text{BatSpeedSSMax} = 72.7$  mph), producing elevated batted ball outcomes ( $\text{ExitVelocity} = 93.7$  mph,  $\text{LaunchAngle} = 24.5^\circ$ ,  $\text{LaunchDirection} = 0.22^\circ$ ). These features suggest reliance on hand speed over lower-body drive.

**Cluster 3: Compact Mechanics with Early Bat Path Establishment** Batters in Cluster 3 demonstrated the most compact mechanical profiles, characterized by the shortest stride length ( $\text{StrideLengthHeelStrike} = 3.11$  ft) and moderate knee flexion ( $\text{FrontKneeFlexionSetUp} = 43.3^\circ$ ,  $\text{RearKneeFlexionSetUp} = 40.1^\circ$ ). Hip-shoulder separation at setup was the lowest ( $\text{HSSeparationSetUp} = -6.8^\circ$ ), and separation reversed at contact ( $\text{HSSeparationContact} = -2.1^\circ$ ), indicating inefficient rotational timing. However, this group achieved the earliest bat entry into the zone ( $\text{BackZoneDepth} = -0.71$  ft) and maintained an aggressive uphill bat path ( $\text{AttackAngleVert} = +3.21^\circ$ ). Forward bend was minimal at contact ( $\text{ForwardBendContact} = 2.5^\circ$ ). Despite mechanical inefficiencies, hitters generated the highest median exit velocity ( $\text{ExitVelocity} = 94.5$  mph) with a line-drive launch profile ( $\text{LaunchAngle} = 18.25^\circ$ ,  $\text{LaunchDirection} = 0.95^\circ$ ), supported by strong bat speed ( $\text{BatSpeedSSMax} = 71.4$  mph). These results suggest compact swings focused on early plane establishment and quick barrel delivery.

**Cluster 4: Deep Load with Late Bat Entry and Downward Attack Path** Cluster 4 hitters exhibited substantial lower-body loading, with the deepest knee flexion among groups ( $\text{FrontKneeFlexionSetUp} = 55.6^\circ$ ,  $\text{RearKneeFlexionSetUp} = 60.0^\circ$ ) and a stride length of 3.24 ft. Hip-shoulder separation was moderate at setup ( $\text{HSSeparationSetUp} = -8.5^\circ$ ) and increased modestly at contact ( $\text{HSSeparationContact} = +3.1^\circ$ ). Forward bend was high both at setup ( $\text{ForwardBendSetUp} = 16.6^\circ$ ) and contact ( $\text{ForwardBendContact} = 7.5^\circ$ ), indicating a more crouched posture throughout the swing. Unlike other clusters, these hitters entered the hitting zone late ( $\text{BackZoneDepth} = +0.08$  ft) and displayed a downward bat path ( $\text{AttackAngleVert} = -1.08^\circ$ ). Bat speed was the lowest ( $\text{BatSpeedSSMax} = 69.1$  mph), with solid but lower batted ball outcomes ( $\text{ExitVelocity} = 93.5$  mph,  $\text{LaunchAngle} = 10^\circ$ ,  $\text{LaunchDirection} = 4.13^\circ$ ). These mechanical patterns reflect a swing structure emphasizing deep loading and downward barrel delivery, with a timing-based approach rather than maximal power output.

To evaluate whether the identified mechanical clusters corresponded to differences in performance, we separately examined swing outcomes (swing-and-miss rates) and hit outcomes (types of batted ball results) across clusters. As shown in Figures 8 and 9, there were no substantial differences among clusters in either swing or hit performance metrics. The proportion of swings resulting in hits was relatively consistent, ranging from 81% to 85%, while swing-and-miss rates varied only slightly between 15% and 19%. Similarly, the distribution of hit types—singles, extra-base hits, fouls, and outs—remained stable across clusters. These findings suggest that although biomechanical swing patterns differed meaningfully, immediate performance outcomes were largely comparable among the groups.

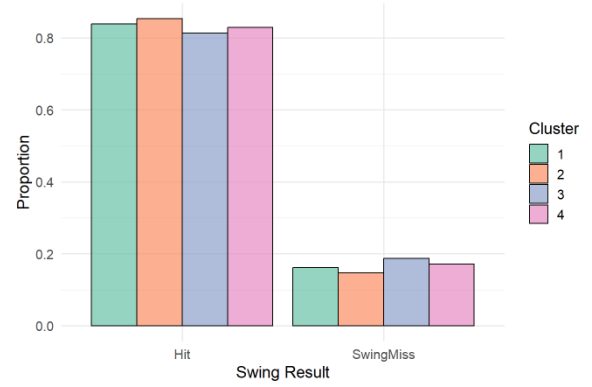


Fig. 8: Swing outcomes (hits vs. swing-and-miss rates) across the representative sets of each cluster.

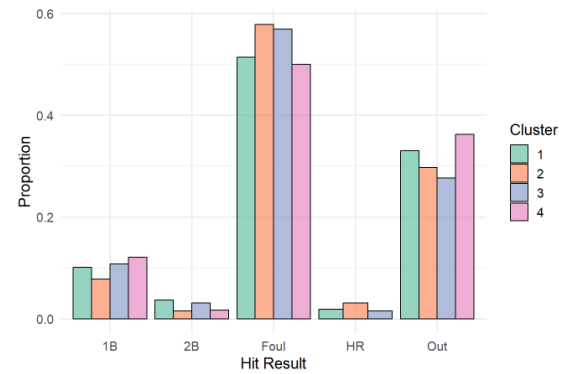


Fig. 9: Distribution of hit types (singles, extra-base hits, fouls, and outs) across the representative sets of each cluster.

#### IV. DISCUSSION

The biomechanical clusters uncovered in this study reflect distinct swing architectures that may correspond to different offensive roles, mechanical risk profiles, and coaching interventions. Cluster 1 batters demonstrated strong lower-body engagement, characterized by deep knee flexion, moderate hip-shoulder separation, and early bat plane establishment. Their balanced movement profiles and efficient rotational sequencing suggest hitters who generate force through controlled lower-body loading and kinetic chain efficiency, offering a blend of gap power and mechanical repeatability suitable for versatile lineup roles.

Cluster 2 featured hitters with more upright postures, reduced knee flexion, and elevated reliance on explosive hand speed. Despite achieving the highest bat speed and early zone entry, minimal hip-shoulder separation at contact suggests a less efficient transfer of energy from the lower body. These profiles align with compact, quick-swing hitters whose mechanical structures favor fast hands over rotational buildup, potentially supporting contact-oriented or aggressive situational hitting strategies.

Cluster 3 exhibited the most compact mechanical profiles, with the shortest stride lengths, earliest bat path establishment,

and steepest uphill attack angles. While demonstrating the highest median exit velocities, these hitters displayed poor rotational timing, reflected in limited or negative hip–shoulder separation at contact. This swing architecture suggests a high-risk, high-reward profile built around early barrel entry and quickness rather than optimized energy transfer, making them potent but timing-sensitive hitters.

In contrast, Cluster 4 emphasized deep lower-body loading, late bat entry, and a downward attack angle. These hitters maintained moderate hip–shoulder separation, pronounced forward bend, and a controlled, timing-based swing structure. Their profiles are consistent with contact-oriented hitters who prioritize bat control, ground-ball or low line-drive trajectories, and consistent barrel contact over maximum exit velocity, aligning with offensive roles that emphasize batting average, table-setting, and situational adaptability.

Figures 8 and 9 show that biomechanical archetypes derived from full-body swing motion do not directly correspond to large differences in immediate outcomes, though the statistical power of this analysis is weakened by the relatively small sizes of the representative sets. However, the distinct mechanical strategies observed across clusters may have long-term implications for player development, efficiency, and risk of breakdown—warranting further exploration.

Taken together, these clusters offer a lens into how movement strategies map onto player roles and physical demands. From a coaching standpoint, the findings can inform individualized training plans—reinforcing desirable patterns, targeting inefficiencies, or designing injury prevention protocols based on biomechanical tendencies.

Our analysis of professional baseball players’ swings identified four distinct biomechanical clusters that differentiate how hitters load, sequence, and deliver their swings. These groupings reflect key movement strategies, including lower-body driven loading with efficient rotational sequencing, upright postural setups emphasizing hand speed, compact swing architectures focused on early bat path establishment, and deep loading patterns paired with timing-based barrel delivery. While exploratory runs of the 250 UMAP and GMM iterations occasionally favored six or seven clusters based on the BIC from the Mclust() function, we opted for a four-cluster solution to balance interpretability and parsimony. Figure 5 supports this decision, as meaningful substructure is visible at lower dendrogram heights. For users seeking more granular segmentation, Table II and the dendrogram highlight that Cluster 3 is the most internally cohesive, while Clusters 1, 2, and 4 could each be split into more specific subgroups with increased robustness. These additional splits could yield useful insights, but may also reduce the number of representative batters available for downstream analysis.

Unlike prior studies that primarily focus on bat path metrics such as barrel speed, attack angle, or swing kinematics derived from external sensors and bat trackers [1], [2], [4], [5], [7], our analysis centers on full-body biomechanics to distinguish between swing types. By leveraging the richness of joint-level tracking data collected at high resolution, we were able to

uncover nuanced movement patterns that differentiate hitters based on how their bodies move throughout the swing — not just how the bat moves. Importantly, our application of Functional Principal Component Analysis (FPCA) allowed us to treat entire joint trajectories as structured functional objects rather than isolated points in time, providing a more holistic view of mechanical variation across swings [3], [6]. It’s worth noting, however, that our results are based on a controlled subset of swings — specifically, right-handed batters facing fastballs near the center of the strike zone. As such, comparisons to broader swing studies should be made with caution.

However, our study has several limitations. First, UMAP introduces a known reproducibility issue: even with fixed random seeds, the results can vary across machines due to platform-dependent randomness. While our use of consensus clustering—repeating UMAP and GMM 250 times—helps stabilize these outcomes, it does not fully resolve this issue and prevents exact replication of our results.

Second, applying our model to new batters poses a challenge. Because UMAP does not naturally allow new data points to be embedded into an existing low-dimensional space, we cannot assign new batters to clusters using the same consensus framework. To overcome this, we trained a random forest on summary biomechanical features to assign new observations to the most appropriate cluster. This introduces a slight methodological shift, but it offers a practical solution for out-of-sample inference.

We also used this same set of summary features to interpret each cluster. While this may seem like a limitation compared to using the full functional data, it was an intentional design choice. Summary statistics offer more interpretable and actionable insights for practitioners, such as coaches and analysts, while still reflecting the core differences captured during clustering.

Finally, although GMMs offer probabilistic cluster membership, our consensus approach relies on hard assignments, which means we lose the ability to represent uncertainty in cluster boundaries—particularly for swing types that may lie near the margins of multiple groups.

Therefore, for future research, we recommend using kernel principal component analysis (kernel PCA) rather than UMAP. Kernel PCA is another nonlinear dimension reduction technique, which is much more computationally expensive than UMAP, but provides several key advantages for our purposes. It is non-stochastic, meaning that a study done with kernel PCA would be reproducible. It also provides loadings to apply to new data points, meaning that a model could assign new data points to clusters in the same way that the original data points were assigned, and the model would be able to be interpreted using the same model that originally clustered the batters. The non-stochastic property of kernel PCA also means that consensus clustering would be unnecessary, so we would be able to retain the nuance of the probabilistic assignments of our GMM.

We also recommend that future research broaden the scope

beyond the narrow context of this study, which focused solely on right-handed batters facing fastballs located near the center of the strike zone. An initial step could involve investigating whether batter biomechanics differ meaningfully based on handedness or pitch characteristics such as type and location. Depending on the outcome, subsequent studies could either specialize in distinct batting scenarios or work toward a more generalized understanding of swing mechanics. In either case, expanding the scope would provide teams with more comprehensive insights into what constitutes an effective swing across a wider range of game contexts.

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## **Technical Appendix**

**TABLE VI:** PCA Loadings for Each Joint on X, Y, and Z Axes

| <b>Joint</b>  | <b>X</b> | <b>Y</b> | <b>Z</b> |
|---------------|----------|----------|----------|
| Hips          | 0.6666   | -0.6794  | 0.3067   |
| LeftAnkle     | 0.3296   | -0.7299  | -0.5987  |
| LeftElbow     | -0.6348  | 0.2272   | 0.7349   |
| LeftHip       | 0.6492   | -0.3203  | -0.4359  |
| LeftKnee      | 0.5164   | -0.6052  | -0.6058  |
| LeftShoulder  | 0.2767   | -0.6645  | -0.6942  |
| LeftWrist     | 0.7060   | -0.2894  | -0.6462  |
| MidShoulder   | 0.4544   | -0.6480  | 0.6112   |
| RightAnkle    | 0.7038   | 0.7099   | 0.0028   |
| RightElbow    | 0.5722   | -0.6247  | 0.5313   |
| RightKnee     | 0.6959   | -0.6090  | 0.3835   |
| RightHip      | 0.6155   | -0.6022  | 0.5084   |
| RightShoulder | 0.3830   | -0.5858  | 0.5921   |
| RightWrist    | 0.7864   | -0.4556  | -0.4171  |

| Batter Label | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Batter Label | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 |
|--------------|-----------|-----------|-----------|-----------|--------------|-----------|-----------|-----------|-----------|
| 1            | 10        | 0         | 0         | 0         | 51           | 3         | 3         | 4         | 0         |
| 2            | 1         | 9         | 0         | 0         | 52           | 10        | 0         | 0         | 0         |
| 3            | 9         | 1         | 0         | 0         | 53           | 4         | 0         | 6         | 0         |
| 4            | 3         | 5         | 2         | 0         | 54           | 7         | 1         | 2         | 0         |
| 5            | 0         | 1         | 9         | 0         | 55           | 0         | 3         | 7         | 0         |
| 6            | 0         | 0         | 0         | 10        | 56           | 0         | 3         | 7         | 0         |
| 7            | 0         | 1         | 9         | 0         | 57           | 3         | 0         | 7         | 0         |
| 8            | 6         | 0         | 4         | 0         | 58           | 0         | 10        | 0         | 0         |
| 9            | 7         | 3         | 0         | 0         | 59           | 10        | 0         | 0         | 0         |
| 10           | 7         | 1         | 2         | 0         | 60           | 0         | 3         | 7         | 0         |
| 11           | 0         | 0         | 10        | 0         | 61           | 7         | 2         | 1         | 0         |
| 12           | 6         | 4         | 0         | 0         | 62           | 1         | 9         | 0         | 0         |
| 13           | 9         | 1         | 0         | 0         | 63           | 10        | 0         | 0         | 0         |
| 14           | 8         | 2         | 0         | 0         | 64           | 1         | 4         | 5         | 0         |
| 15           | 7         | 1         | 2         | 0         | 65           | 0         | 10        | 0         | 0         |
| 16           | 9         | 1         | 0         | 0         | 66           | 0         | 6         | 4         | 0         |
| 17           | 0         | 1         | 9         | 0         | 67           | 10        | 0         | 0         | 0         |
| 18           | 0         | 10        | 0         | 0         | 68           | 0         | 4         | 6         | 0         |
| 19           | 10        | 0         | 0         | 0         | 69           | 3         | 0         | 7         | 0         |
| 20           | 4         | 0         | 6         | 0         | 70           | 0         | 10        | 0         | 0         |
| 21           | 0         | 3         | 7         | 0         | 71           | 6         | 0         | 4         | 0         |
| 22           | 9         | 1         | 0         | 0         | 72           | 9         | 0         | 1         | 0         |
| 23           | 0         | 0         | 10        | 0         | 73           | 0         | 8         | 2         | 0         |
| 24           | 0         | 7         | 3         | 0         | 74           | 1         | 9         | 0         | 0         |
| 25           | 0         | 0         | 10        | 0         | 75           | 6         | 4         | 0         | 0         |
| 26           | 0         | 3         | 7         | 0         | 76           | 3         | 0         | 7         | 0         |
| 27           | 0         | 6         | 4         | 0         | 77           | 10        | 0         | 0         | 0         |
| 28           | 0         | 10        | 0         | 0         | 78           | 6         | 4         | 0         | 0         |
| 29           | 1         | 9         | 0         | 0         | 79           | 4         | 0         | 6         | 0         |
| 30           | 0         | 1         | 9         | 0         | 80           | 7         | 3         | 0         | 0         |
| 31           | 0         | 0         | 10        | 0         | 81           | 7         | 0         | 3         | 0         |
| 32           | 10        | 0         | 0         | 0         | 82           | 0         | 8         | 2         | 0         |
| 33           | 8         | 2         | 0         | 0         | 83           | 0         | 4         | 6         | 0         |
| 34           | 7         | 1         | 2         | 0         | 84           | 9         | 1         | 0         | 0         |
| 35           | 0         | 5         | 5         | 0         | 85           | 0         | 0         | 10        | 0         |
| 36           | 4         | 3         | 3         | 0         | 86           | 0         | 8         | 2         | 0         |
| 37           | 5         | 5         | 0         | 0         | 87           | 10        | 0         | 0         | 0         |
| 38           | 0         | 0         | 10        | 0         | 88           | 10        | 0         | 0         | 0         |
| 39           | 1         | 0         | 9         | 0         | 89           | 1         | 9         | 0         | 0         |
| 40           | 0         | 0         | 10        | 0         | 90           | 0         | 6         | 4         | 0         |
| 41           | 0         | 0         | 10        | 0         | 91           | 4         | 5         | 1         | 0         |
| 42           | 3         | 5         | 2         | 0         | 92           | 0         | 10        | 0         | 0         |
| 43           | 5         | 5         | 0         | 0         | 93           | 3         | 7         | 0         | 0         |
| 44           | 0         | 10        | 0         | 0         | 94           | 9         | 1         | 0         | 0         |
| 45           | 0         | 3         | 7         | 0         | 95           | 7         | 3         | 0         | 0         |
| 46           | 4         | 6         | 0         | 0         | 96           | 0         | 0         | 10        | 0         |
| 47           | 0         | 10        | 0         | 0         | 97           | 1         | 9         | 0         | 0         |
| 48           | 2         | 3         | 5         | 0         | 98           | 0         | 0         | 10        | 0         |
| 49           | 6         | 4         | 0         | 0         | 99           | 8         | 0         | 2         | 0         |
| 50           | 8         | 2         | 0         | 0         | 100          | 0         | 7         | 3         | 0         |

**TABLE VII:** Swing Consistency: Number of Swings per Cluster for Each Batter (split across two columns).



| Feature                         | Cluster 1        | Cluster 2        | Cluster 3        | Cluster 4        |
|---------------------------------|------------------|------------------|------------------|------------------|
| TimeSetUp                       | (-0.37, -0.26)   | (-0.53, -0.39)   | (-0.63, -0.47)   | (-0.37, -0.28)   |
| TimePeakGather                  | (-0.13, -0.06)   | (-0.34, -0.23)   | (-0.43, -0.29)   | (-0.24, -0.19)   |
| StrideLengthSetUp (ft)          | (2.37, 2.84)     | (2.03, 2.49)     | (1.72, 2.19)     | (2.16, 2.42)     |
| StrideLengthHeelStrike (ft)     | (3.36, 3.45)     | (3.09, 3.44)     | (3.03, 3.28)     | (2.94, 3.49)     |
| StrideLengthContact (ft)        | (3.04, 3.24)     | (2.92, 3.14)     | (2.76, 2.98)     | (2.80, 3.14)     |
| StrideWidthSetUp (ft)           | (0.10, 0.28)     | (0.05, 0.61)     | (0.07, 0.58)     | (0.14, 0.29)     |
| StrideWidthContact (ft)         | (-0.11, 0.33)    | (-0.11, 0.19)    | (0.05, 0.37)     | (-0.17, 0.05)    |
| FrontKneeFlexionSetUp (deg)     | (40.91, 60.72)   | (29.99, 39.03)   | (38.32, 49.45)   | (50.25, 57.48)   |
| RearKneeFlexionSetUp (deg)      | (53.94, 64.02)   | (34.06, 48.67)   | (38.57, 52.56)   | (59.37, 61.38)   |
| RearKneeFlexionPeakGather (deg) | (58.42, 71.90)   | (36.90, 49.74)   | (38.60, 57.98)   | (61.98, 66.19)   |
| RearKneeFlexionHeelStrike (deg) | (56.66, 66.74)   | (51.35, 57.87)   | (54.40, 64.59)   | (63.66, 67.43)   |
| HipPlateDistContact (ft)        | (2.94, 3.06)     | (3.17, 3.28)     | (2.95, 3.07)     | (2.87, 3.03)     |
| HipsXSetUp (ft)                 | (-1.15, -0.86)   | (-1.31, -1.12)   | (-1.66, -1.43)   | (-0.70, -0.34)   |
| HipsXHeelStrike (ft)            | (-0.34, -0.04)   | (-0.21, 0.02)    | (-0.47, -0.30)   | (0.25, 0.43)     |
| HipsXContact (ft)               | (-0.37, -0.06)   | (-0.20, 0.01)    | (-0.52, -0.27)   | (0.17, 0.39)     |
| HSSeparationSetUp (deg)         | (-12.93, -6.03)  | (-19.19, -7.25)  | (-13.32, -2.56)  | (-9.69, -3.22)   |
| HSSeparationContact (deg)       | (4.36, 8.40)     | (-3.98, 2.60)    | (-4.48, -0.26)   | (0.89, 9.18)     |
| ForwardBendSetUp (deg)          | (13.07, 20.10)   | (5.54, 13.15)    | (7.56, 14.60)    | (16.68, 19.50)   |
| ForwardBendPeakGather (deg)     | (18.14, 22.28)   | (7.81, 16.56)    | (10.29, 16.14)   | (17.79, 20.82)   |
| ForwardBendHeelStrike (deg)     | (20.62, 22.72)   | (16.65, 20.89)   | (15.64, 20.51)   | (16.88, 21.13)   |
| ForwardBendContact (deg)        | (4.19, 8.65)     | (3.71, 6.72)     | (0.85, 4.06)     | (3.33, 7.37)     |
| TorsoRotationHeelStrike (deg)   | (-79.48, -70.54) | (-72.87, -68.98) | (-72.78, -60.87) | (-71.72, -60.72) |
| FrontZoneDepth (ft)             | (2.30, 2.61)     | (2.34, 2.70)     | (2.10, 2.50)     | (2.90, 2.97)     |
| BackZoneDepth (ft)              | (-0.65, -0.42)   | (-0.65, -0.46)   | (-0.91, -0.53)   | (-0.11, 0.12)    |

**TABLE VIII:** Non-normalized interquartile ranges of biomechanical features used for cluster profiling.